

Notes: Dávila and Parlatore (RFS, 2026)

Identifying Price Informativeness

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Main data sources	4
2.2	Main empirical variables	4
2.3	Stock characteristics used in the panel regressions	5
2.4	Observed versus unobserved objects	5
3	General framework (all equations + interpretation)	5
3.1	Payoff process	5
3.2	Equilibrium pricing equation	6
3.3	The price as an endogenous signal	6
3.4	Absolute price informativeness	6
3.5	Relative price informativeness	7
3.6	Bayesian updating interpretation	7
4	Identification and estimation (what makes the estimates credible)	7
4.1	Proposition 1: baseline identification	7
4.2	Why the formula is nontrivial	8
4.3	Unlearnable payoff component	8
4.4	Longer horizons	9
4.5	Panel implementation	9
4.6	What makes the estimates credible	10
4.7	Main limitations of identification	10
5	Structural models (economic interpretation)	10
5.1	Model 1: sentiment as noise	11

5.2	Model 2: representative agent	11
5.3	Model 3: informed, uninformed, and noise traders	11
5.4	Why the structural models matter	12
6	Tables and figures	12
6.1	Figure 1: Interpreting relative price informativeness	12
6.2	Table 1: Baseline price informativeness summary statistics	13
6.3	Figure 2: Relative-frequency histogram	13
6.4	Table 2: Cross-sectional results	13
6.5	Figures 3 and 4: Exchange, S&P 500 status, and industry	14
6.6	Figure 5: Price informativeness over time	15
6.7	Figure 6: Time trends by characteristics	15
6.8	Figures 7 and 8: Unlearnable component results	16
6.9	Table 3: Longer horizons	17
7	Relation to existing informativeness measures	17
7.1	Posterior variance and forecasting price efficiency	17
7.2	Informativeness versus predictability	18
8	Limitations and open issues	18
9	Short summary	18
10	Conclusion	19

1 Big picture

- **Question.** How can researchers identify and estimate how informative asset prices are about future payoffs? The paper is about turning the theoretical object “price informativeness” into an empirically measurable object.
- **Motivation.** Many theories in finance treat prices as aggregators of dispersed information. But empirical work often uses indirect proxies for informativeness. This paper asks: starting from a pricing equation and a payoff process, what exactly can be recovered from price and payoff data?
- **Main theoretical object.** The paper focuses on **relative price informativeness**, denoted τ_π^R . It is the precision of the price signal relative to total precision. It lies between 0 and 1.
- **Interpretation.** If $\tau_\pi^R = 0.05$, then an external observer who learns only from the price reduces uncertainty about the future payoff innovation by 5%. In a Gaussian environment, the same number is the Kalman gain: the observer puts 5% weight on the price signal and 95% weight on the prior.
- **Core identification result.** Relative price informativeness is recovered from two regressions of price changes on payoff changes. The key formula is

$$\tau_\pi^R = \frac{R_{\Delta x, \Delta x'}^2 - R_{\Delta x}^2}{1 - R_{\Delta x}^2}.$$

The numerator is the extra explanatory power from future payoff growth; the denominator is residual uncertainty after conditioning on current payoff growth.

- **Why this is powerful.** The econometrician does *not* need to observe the source of noise in prices. Noise can come from sentiment, risk-bearing capacity, time-varying risk premia, private trading needs, or noise-trader demand. The method only requires price data, payoff data, and whatever public controls the observer wants to condition on.
- **Empirical application.** The paper estimates a quarterly panel of stock-specific relative price informativeness for U.S. stocks. The main data are CRSP/Compustat prices and accounting variables, plus IBES analyst coverage and forecasts for some extensions.
- **Main empirical finding 1: level.** In the baseline implementation, price informativeness is small but economically meaningful. The time-series averages of the cross-sectional median and mean are 2.34% and 2.61%, respectively.
- **Main empirical finding 2: cross-section.** Price informativeness is higher for larger stocks, high-turnover stocks, stocks with higher institutional ownership, and stocks with more analyst coverage. Value stocks have lower informativeness. Once size is controlled for, institutional ownership loses its positive relation and analyst coverage becomes much weaker.
- **Main empirical finding 3: time series.** Mean, median, and dispersion of informativeness increase from the mid-1980s to 2017. The median rises from roughly 1.66% to 2.95%, and the mean from roughly 1.72% to 3.29%.
- **Main extension.** When the payoff has an unlearnable component and analyst forecasts proxy for the learnable component, estimated informativeness is lower: the average median and mean from 1996 to 2017 are 0.47% and 0.85%.

- **Main contribution.** The paper provides a bridge between information aggregation theory and empirical measurement. It gives a model-based way to test theories about which assets have more informative prices and how informativeness changes over time.
- **Important caveat.** The results are exact inside a linear or log-linear environment. The empirical implementation also depends on how payoffs are measured, how public controls are specified, and whether earnings growth is a good object for future fundamentals.

2 Data and measurement (key objects)

2.1 Main data sources

- **Price and stock data.** The empirical analysis uses CRSP data for prices, market capitalization, turnover, S&P 500 membership, exchange, and industry classification.
- **Accounting data.** Compustat provides earnings and book values at quarterly and annual frequency. Nominal variables are deflated using the PCE price index.
- **Payoff proxy.** The payoff variable is earnings, measured using EBIT. The authors use earnings rather than dividends because firm-level dividends are often zero, lumpy, or not a clean measure of fundamentals.
- **Analyst data.** IBES provides analyst coverage and, in the unlearnable-payoff extension, four-quarter-ahead analyst forecasts of earnings growth.
- **Sample period.** The raw quarterly data are available from 1966 to 2017. The main time-series figures start in 1985 because earlier samples are small. The unlearnable-payoff implementation runs mainly from 1996 to 2017.
- **Stock universe.** The baseline estimates produce a panel for 4,920 unique stocks after dropping negative informativeness estimates. On average, more than 93% of stocks have positive estimated informativeness in a given year.

2.2 Main empirical variables

- p_t^j : log price of stock j , adjusted for splits.
- Δp_t^j : year-over-year log price change, interpreted as capital gains.
- x_t^j : payoff proxy for stock j , based on earnings.
- Δx_t^j : year-over-year earnings growth, scaled by book equity because earnings can be negative.
- Δx_{t+4}^j : one-year-ahead earnings growth. This is the future payoff growth whose information content in prices is being measured.
- Y_t^j : vector of stock characteristics and controls. It includes variables used to capture asset demand and information acquisition differences across stocks.

2.3 Stock characteristics used in the panel regressions

The empirical implementation lets regression coefficients and residual variances vary with stock characteristics. The main characteristics are:

- **Size:** log market capitalization.
- **Value:** book-to-market ratio.
- **Turnover:** trading volume divided by shares outstanding.
- **Idiosyncratic volatility:** standard deviation over 30 months of stock return minus market return.
- **Institutional ownership:** fraction of shares held by institutional investors.
- **Analyst coverage:** number of IBES analysts covering the stock.

The vector Y_t^j also includes changes in profitability, dividend ratio, asset growth, and market beta, following the asset-demand controls in Kojien and Yogo-style specifications.

Interpretation. The authors do not run separate time-series regressions for each stock because that would be very noisy. Instead, they use a panel structure that lets the relevant coefficients and variances depend on stock observables.

2.4 Observed versus unobserved objects

- **Observed:** prices, earnings, stock characteristics, ownership, analyst coverage, and analyst forecasts.
- **Unobserved:** the private signals investors receive, the noise in prices, the source of non-payoff trading motives, and the exact information set of market participants.
- **Key methodological point.** The paper's identification result is valuable because price informativeness can be recovered without observing the unobserved noise term.

3 General framework (all equations + interpretation)

3.1 Payoff process

The risky asset has log payoff x_{t+1} . Payoff growth follows a difference-stationary AR(1):

$$\Delta x_{t+1} = \mu_{\Delta x} + \rho \Delta x_t + u_t, \quad |\rho| < 1. \quad (1)$$

Interpretation.

- u_t is the innovation to next-period payoff growth.
- The index t is intentional: investors may learn about u_t at date t before the payoff is realized.
- The empirical version maps x_t to earnings rather than dividends.

3.2 Equilibrium pricing equation

The equilibrium log price difference is assumed to be linear:

$$\Delta p_t = \phi + \phi_0 \Delta x_t + \phi_1 \Delta x_{t+1} + \phi'_\chi \Delta \chi_t + \phi_n \Delta n_t. \quad (2)$$

The public signal vector is

$$\Delta \chi_t = \omega_\chi u_t + \varepsilon_t^\chi,$$

and non-payoff trading motives satisfy

$$\Delta n_t = \mu_{\Delta n} + \omega_n u_t + \varepsilon_t^{\Delta n}.$$

Interpretation.

- $\phi_1 \Delta x_{t+1}$ captures the part of prices driven by information about future payoffs.
- $\phi_n \Delta n_t$ captures trading motives not directly based on information, such as sentiment, risk-bearing capacity, or noise trading.
- The model allows non-payoff trading motives to be correlated with payoff innovations through $\omega_n u_t$.
- The equation is not tied to one particular microfoundation. Section 2 of the paper shows that several structural models generate this kind of linear/log-linear price equation.

3.3 The price as an endogenous signal

The price can be transformed into an unbiased signal about the payoff innovation u_t :

$$\pi_t \equiv \frac{1}{\phi_1 + \phi_n \omega_n} \left(\Delta p_t - \left[\phi + \phi_1 \mu_{\Delta x} + \phi_n \mu_{\Delta n} + (\phi_0 + \phi_1 \rho) \Delta x_t + \phi'_\chi \Delta \chi_t \right] \right). \quad (1)$$

In the paper's notation, this signal can be written as

$$\pi_t = u_t + \frac{\phi_n}{\phi_1 + \phi_n \omega_n} (\Delta n_t - \mu_{\Delta n}). \quad (3)$$

Interpretation.

- The price is informative because it loads on the future payoff innovation u_t .
- The price is noisy because it also loads on non-payoff trading motives.
- The econometrician does not need to observe Δn_t to identify relative informativeness.

3.4 Absolute price informativeness

Absolute price informativeness is the precision of the price signal about u_t :

$$\tau_\pi \equiv (\text{Var}[\pi_t \mid \Delta x_{t+1}, \Delta x_t, \Delta \chi_t])^{-1} = \left(\frac{\phi_1 + \phi_n \omega_n}{\phi_n} \right)^2 \tau_{\Delta n}, \quad (4)$$

where $\tau_{\Delta n} = \text{Var}[\Delta n_t]^{-1}$.

Interpretation.

- Higher loading on future payoffs raises informativeness.
- More noise from non-payoff trading motives lowers informativeness.
- Absolute informativeness has units, so it is hard to compare across stocks with different payoff volatility.

3.5 Relative price informativeness

Relative price informativeness normalizes the precision of the price signal by total precision:

$$\tau_{\pi}^R \equiv \frac{\tau_{\pi}}{\tau_{\pi} + \tau_{u|\Delta\chi_t}}, \quad \tau_{u|\Delta\chi_t} \equiv \text{Var}[u_t | \Delta x_t, \Delta\chi_t]^{-1}. \quad (5)$$

Interpretation.

- $\tau_{\pi}^R \in [0, 1]$.
- It is unit-free, so it can be compared across stocks and across time.
- It measures how much uncertainty about the payoff innovation is eliminated by observing the price.
- In a Gaussian setting, it is exactly the Bayesian Kalman gain on the price signal.

3.6 Bayesian updating interpretation

If the variables are Gaussian, an external observer who only learns from the price has posterior

$$u_t | \Delta p_t, \Delta x_t, \Delta\chi_t \sim N\left(\tau_{\pi}^R \pi_t, (\tau_{\pi} + \tau_u)^{-1}\right). \quad (7)$$

Interpretation.

- τ_{π}^R is the weight put on the price signal.
- $1 - \tau_{\pi}^R$ is the weight put on the prior.
- A price informativeness estimate of 2.5% means that the price signal is useful, but the observer still places most weight on prior/payoff information.

4 Identification and estimation (what makes the estimates credible)

4.1 Proposition 1: baseline identification

The baseline identification result uses two regressions.

First, regress price growth on current and future payoff growth plus public information:

$$\Delta p_t = \beta + \beta_0 \Delta x_t + \beta_1 \Delta x_{t+1} + \beta_2' \Delta\chi_t + e_t. \quad (\text{R1})$$

Let the R^2 from this regression be $R_{\Delta x, \Delta x'}^2$.

Second, regress price growth only on current payoff growth plus public information:

$$\Delta p_t = \zeta + \zeta_0 \Delta x_t + \zeta'_2 \Delta \chi_t + e_t^\zeta. \quad (\text{R2})$$

Let the R^2 from this regression be $R_{\Delta x}^2$.

Then relative price informativeness is

$$\tau_\pi^R = \frac{R_{\Delta x, \Delta x'}^2 - R_{\Delta x}^2}{1 - R_{\Delta x}^2}. \quad (6)$$

Interpretation.

- $R_{\Delta x}^2$ captures how much of price growth is explained by current payoff growth and public information.
- $R_{\Delta x, \Delta x'}^2 - R_{\Delta x}^2$ captures the extra explanatory power of future payoff growth.
- $1 - R_{\Delta x}^2$ is the residual uncertainty after conditioning on current payoff growth and public information.
- The ratio is therefore the reduction in residual uncertainty generated by the information about future payoffs contained in prices.

4.2 Why the formula is nontrivial

- The econometrician does not observe Δn_t , the non-payoff component of prices.
- In many models, price noise is endogenous and may be correlated with payoffs.
- The proposition shows that a specific combination of R^2 terms recovers the theoretical object despite not observing the noise.
- OLS consistently estimates the relevant R^2 objects under the model assumptions.

4.3 Unlearnable payoff component

The baseline model assumes all payoff innovation can be learned. The paper also considers the case in which

$$u_t = u_t^L + u_t^U, \quad (2)$$

where u_t^L is learnable and u_t^U is unlearnable. Investors receive private signals only about u_t^L .

The price process becomes

$$\Delta p_t = \phi + \phi_0 \Delta x_t + \phi_1 \Delta u_t^L + \phi'_\chi \Delta \chi_t + \phi_n \Delta n_t.$$

Relative informativeness about the learnable component is

$$\tau_\pi^{RL} = \frac{\tau_\pi^L}{\tau_{u_t^L | \Delta \chi_t} + \tau_\pi^L}. \quad (9)$$

The corresponding identification formula is

$$\tau_{\pi}^{RL} = \frac{R_{u,u'}^2 - R_u^2}{1 - R_u^2}. \quad (10)$$

Empirical mapping.

- The paper uses analyst forecasts of future earnings growth to proxy for the learnable component of the payoff.
- This is conceptually appealing because analysts are explicitly trying to forecast earnings.
- It is also imperfect because forecasts may mechanically react to prices.

4.4 Longer horizons

The price can contain information about payoff growth at horizons longer than one period:

$$\Delta p_t = \phi + \phi_0 \Delta x_t + \sum_{m=1}^M \phi_m \Delta x_t^m + \phi'_\chi \Delta \chi_t + \phi_n \Delta n_t, \quad (11)$$

where

$$\Delta x_t^m \equiv x_{t+m} - x_t.$$

For horizon m , the identification formula is

$$\tau_{\pi}^{mR} = \frac{R_{\Delta x, \Delta x^m}^2 - R_{\Delta x}^2}{1 - R_{\Delta x}^2}. \quad (12)$$

Interpretation.

- The same logic works for multi-year payoff growth.
- Empirically, informativeness tends to fall as the horizon grows.
- This is intuitive: prices are usually more informative about nearer-term fundamentals than about far-future fundamentals.

4.5 Panel implementation

The empirical implementation uses panel regressions that make coefficients affine functions of stock characteristics Y_t^j :

$$\Delta p_t^j = \beta(Y_t^j) + \beta_0(Y_t^j) \Delta x_t^j + \beta_1(Y_t^j) \Delta x_{t+4}^j + \varepsilon_t^j, \quad (18)$$

$$\Delta p_t^j = \zeta(Y_t^j) + \zeta_0(Y_t^j) \Delta x_t^j + \tilde{\varepsilon}_t^j. \quad (19)$$

The residual variances are modeled as exponential quadratic functions of characteristics:

$$\text{Var}[\varepsilon_t^j] = \exp \left(\lambda_0 + \lambda_1 Y_t^j + Y_t^{j'} \lambda_2 Y_t^j \right), \quad (3)$$

$$\text{Var}[\tilde{\varepsilon}_t^j] = \exp \left(\tilde{\lambda}_0 + \tilde{\lambda}_1 Y_t^j + Y_t^{j'} \tilde{\lambda}_2 Y_t^j \right). \quad (4)$$

Then stock-specific informativeness is estimated as

$$\hat{\tau}_{\pi,t}^{R,j} = \frac{\widehat{\text{Var}}[\tilde{\varepsilon}_t^j] - \widehat{\text{Var}}[\varepsilon_t^j]}{\widehat{\text{Var}}[\tilde{\varepsilon}_t^j]}. \quad (24)$$

Interpretation.

- Regression (18) is the richer regression with future earnings growth.
- Regression (19) is the restricted regression without future earnings growth.
- If adding future earnings growth reduces the residual variance a lot, the price was highly informative about future fundamentals.
- The exponential variance specification ensures positive variances and allows informativeness to vary by stock characteristics.

4.6 What makes the estimates credible

- **Formal identification.** The empirical formula is not an ad hoc proxy. It is derived from the definition of relative price informativeness.
- **Noise agnosticism.** The approach does not require observing whether noise comes from liquidity trading, sentiment, risk premia, or strategic behavior.
- **Unit-free object.** Relative informativeness is scaled by payoff uncertainty, making comparisons across stocks meaningful.
- **Panel structure.** Coefficients vary with stock characteristics, which reduces noise relative to purely stock-level time-series regressions.
- **Robust conceptual extensions.** The paper studies public signals, unlearnable payoff components, and longer horizons.

4.7 Main limitations of identification

- The core result relies on linear or log-linear price equations.
- The framework abstracts from feedback effects in which prices affect fundamentals.
- Earnings are only one possible payoff proxy.
- The learnable component of earnings is not directly observed.
- High-frequency informativeness is hard to estimate because high-frequency payoff measures are rarely available.

5 Structural models (economic interpretation)

The general framework only assumes a payoff process and a linear pricing equation. Section 2 shows that the pricing equation can arise from several microfounded models. This matters because it explains why the identification result is not tied to one narrow model.

5.1 Model 1: sentiment as noise

- The economy has overlapping generations of investors.
- Investors live for two periods and trade a risk-free asset and a risky asset.
- They receive private signals about the payoff innovation u_t .
- They also have random priors or sentiment n_t , which generates non-payoff trading motives.

The equilibrium log price is approximately

$$\Delta p_t \approx \phi + \phi_0 \Delta x_t + \phi_1 \Delta x_{t+1} + \phi_n \Delta n_t.$$

Interpretation.

- A larger price loading on private signals raises informativeness.
- A larger price loading on sentiment lowers informativeness.
- Empirically, large and liquid stocks may be more informative because informed investors acquire and trade on more information about them.

5.2 Model 2: representative agent

- All investors have the same information and the same prior.
- There is no asymmetric information across investors.
- The price can still be informative to an outside observer because it reflects the representative agent's signal and prior.

The approximate price can be written as

$$p_t \approx \frac{\alpha_x}{\alpha_p} x_t + \frac{\alpha_s}{\alpha_p} s_t + \frac{\alpha_n}{\alpha_p} n_t + \frac{\psi}{\alpha_p}.$$

Interpretation.

- The framework is not only about dispersed information.
- It also applies when all investors have the same information but prices transmit that information to outsiders.
- Time-varying risk aversion or risk premia can appear as another source of price noise.

5.3 Model 3: informed, uninformed, and noise traders

- Informed traders observe the payoff innovation.
- Uninformed traders do not observe it and can learn from the price.
- Noise traders generate random demand.

The approximate price is

$$p_t \approx \frac{\alpha_x}{\alpha_p} x_t + \frac{\alpha_s}{\alpha_p} u_t + \frac{\alpha_n}{\alpha_p} n_t + \frac{\psi}{\alpha_p} + \frac{\delta_t}{\alpha_p}.$$

Interpretation.

- The classic Grossman–Stiglitz-style setting also maps into the paper’s pricing equation.
- Noise trader demand prevents prices from being fully revealing.
- The outside observer faces multiple noise sources: noise demand and investor prior/sentiment variation.

5.4 Why the structural models matter

- They show that the same reduced-form price equation can represent different economic mechanisms.
- The empirical measure identifies informativeness, not the underlying source of informativeness.
- To interpret cross-sectional patterns as changes in private information, noise trading, or risk-bearing capacity, one needs a structural model.

6 Tables and figures

6.1 Figure 1: Interpreting relative price informativeness

Read: Figure 1 explains the R^2 formula visually.

- The denominator $1 - R_{\Delta x}^2$ is the uncertainty about future payoffs left after conditioning on current payoff growth.
- The numerator $R_{\Delta x, \Delta x'}^2 - R_{\Delta x}^2$ is the additional reduction in uncertainty from including future payoff growth.
- The ratio is therefore the fraction of residual uncertainty resolved by the price signal.

Why it matters. This figure gives the simplest intuition for the whole paper. Relative price informativeness is not just another R^2 ; it is a normalized incremental R^2 .

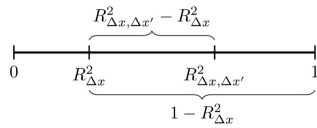


Figure 1
Interpreting relative price informativeness
Relative price informativeness can be computed as the reduction in uncertainty, given by $R_{\Delta x, \Delta x'}^2 - R_{\Delta x}^2$, relative to the remaining residual uncertainty about future payoffs after conditioning on the realized date t payoff, given by $1 - R_{\Delta x}^2$.

Table 1
Price informativeness: Year-over-year summary statistics

t	Median	Mean	SD	Skew	Kurt	P5	P25	P75	P95	n
1985	0.0174	0.0176	0.0080	0.2021	-0.3826	0.0047	0.0116	0.0231	0.0312	1,093
1990	0.0205	0.0212	0.0087	0.3373	0.3908	0.0072	0.0147	0.0272	0.0354	1,163
1995	0.0203	0.0211	0.0130	5.1212	98.2282	0.0039	0.0120	0.0291	0.0391	1,917
2000	0.0221	0.0252	0.0167	1.8982	8.1669	0.0050	0.0137	0.0333	0.0532	1,763
2005	0.0261	0.0295	0.0180	1.6212	5.3952	0.0072	0.0172	0.0380	0.0616	2,027
2010	0.0246	0.0272	0.0178	2.4974	21.2584	0.0055	0.0150	0.0360	0.0557	2,142
2015	0.0290	0.0318	0.0187	1.6314	7.1199	0.0078	0.0190	0.0414	0.0643	1,911

This table reports year-over-year summary statistics (every 5 years) on the panel of price informativeness estimates. It provides information on the median; mean; standard deviation; skewness; excess kurtosis; and 5th, 25th, 75th, and 95th percentiles of each yearly distribution, as well as the number of stocks in each year. This table averages the quarterly measures of price informativeness for a given year before computing the summary statistics.

6.2 Table 1: Baseline price informativeness summary statistics

Read: Table 1 reports the distribution of stock-specific informativeness every five years.

Year	Median	Mean	SD	P95	Number of stocks
1985	0.0174	0.0176	0.0080	0.0312	1,093
1990	0.0205	0.0212	0.0087	0.0354	1,163
1995	0.0203	0.0211	0.0130	0.0391	1,917
2000	0.0221	0.0252	0.0167	0.0532	1,763
2005	0.0261	0.0295	0.0180	0.0616	2,027
2010	0.0246	0.0272	0.0178	0.0557	2,142
2015	0.0290	0.0318	0.0187	0.0643	1,911

Key numbers.

- The time-series average median is 2.34%.
- The time-series average mean is 2.61%.
- The distribution is right-skewed.
- The 95th percentile is usually below about 5–6%.

Interpretation. Most prices contain some information about future earnings, but the price signal is not overwhelmingly precise. Even at the high end of the distribution, the external observer puts only a modest weight on the price signal.

6.3 Figure 2: Relative-frequency histogram

Read: Figure 2 shows the cross-sectional histogram of informativeness in the last quarter of 2015.

- The histogram has 1,752 stocks.
- The distribution is right-skewed.
- Many stocks have low but positive informativeness.

Why it matters. This motivates using the median as a useful summary statistic and motivates cross-sectional analysis: some stocks have much more informative prices than others.

6.4 Table 2: Cross-sectional results

Read: Table 2 regresses average informativeness by twentile bins on stock characteristics, with year fixed effects.

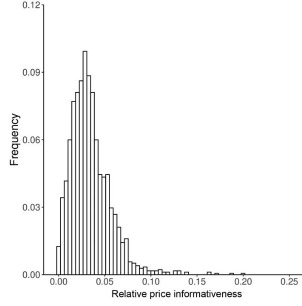


Figure 2
Price informativeness: Relative-frequency histogram
Figure showing a relative-frequency of price informativeness for a representative time period for the last quarter of 2015. The histogram features 1,752 stocks.

Table 2
Cross-sectional results

	Estimate	Std. error	t-stat
Size	0.004880	0.0000706	69.13
Value	-0.013878	0.0004204	-33.01
Turnover	0.000437	0.0000086	51.03
Idiosyncratic volatility	-0.006933	0.0020847	-3.33
Institutional ownership	0.018366	0.0005385	34.11
Analysts covering	0.001499	0.0000201	74.43
Institutional ownership (residualized)	-0.000135	0.0001600	-0.84
Analysts (residualized)	0.000096	0.0000076	12.69

This table reports the estimates ($\hat{a}_i^r$) of panel regressions of price informativeness on cross-sectional characteristics (in twentiles) with year fixed effects ($\hat{c}_t^r$): $\tau_{i,t}^{R,b} = \hat{a}_i^r + \hat{c}_t^r + \epsilon_{i,t}^r$, where $\tau_{i,t}^{R,b}$ denotes the average price informativeness per bin (twentile) in a given period, $\hat{c}_t^r$ denotes the value of the given characteristic per bin (twentile) in a given period, $\hat{a}_i^r$ and $\hat{c}_t^r$ are parameters, and $\epsilon_{i,t}^r$ is an error term. Figures 1A.3 through 1A.7 provide the graphical counterpart of the results in this table. Size is measured as the natural logarithm of stock market capitalization; value is measured as the ratio between a stock's book value and its market capitalization; turnover is measured as the ratio between trading volume and shares outstanding; idiosyncratic volatility is measured as the standard deviation — over a 30-month period — of the difference between the returns of a stock and the market return; institutional ownership is measured as the proportion of a stock held by institutional investors; and analyst coverage measured as the number of analysts covering the stock in a given quarter. The last two rows provide the estimates of the residual of price informativeness on size on institutional ownership and analyst coverage.

Characteristic	Estimate	Std. error	t-stat
Size	0.004880	0.0000706	69.13
Value	-0.013878	0.0004204	-33.01
Turnover	0.000437	0.0000086	51.03
Idiosyncratic volatility	-0.006933	0.0020847	-3.33
Institutional ownership	0.018366	0.0005385	34.11
Analyst coverage	0.001499	0.0000201	74.43
Institutional ownership residualized by size	-0.000135	0.0001600	-0.84
Analysts residualized by size	0.000096	0.0000076	12.69

Interpretation.

- Bigger stocks have more informative prices.
- Higher book-to-market stocks have less informative prices.
- Higher turnover is associated with more informative prices.
- Institutional ownership and analyst coverage are positively related to informativeness before controlling for size.
- After removing the size component, institutional ownership is no longer significant and analyst coverage is much smaller.

Why it matters. The strongest cross-sectional result is that informativeness is concentrated in large, actively traded stocks. This is consistent with information acquisition and trading incentives being stronger where markets are deep and active.

6.5 Figures 3 and 4: Exchange, S&P 500 status, and industry

Read: Figures 3 and 4 compare residual informativeness distributions after removing year fixed effects.

- NYSE stocks have higher median informativeness than NASDAQ stocks.
- AMEX stocks have lower informativeness than the other main exchanges.
- S&P 500 stocks have higher informativeness than non-S&P 500 stocks.

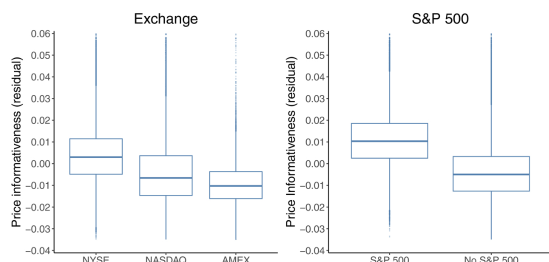


Figure 3
Cross-sectional results
The left panel in this figure shows a box plot by exchange of the residuals of a regression of relative price informativeness on year fixed effects. The right panel in this figure shows a box plot by S&P 500 status of the residuals of a regression of relative price informativeness on year fixed effects. The solid middle line represents the median. The top and bottom of the boxes represent the 75th and 25th percentiles. The whiskers extend up to 1.5 times the interquartile range.

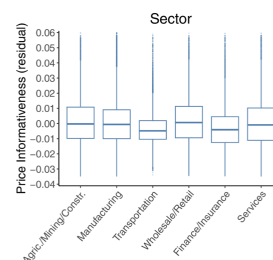


Figure 4
Cross-sectional results
This figure shows a box plot by one-digit SIC industry code of the residuals of a regression of relative price informativeness on year fixed effects. The solid middle line represents the median. The top and bottom of the boxes represent the 75th and 25th percentiles. The whiskers extend up to 1.5 times the interquartile range.

- Manufacturing, wholesale/retail, and services have relatively high median informativeness.
- Transportation and finance have relatively low median informativeness.

Why it matters. These patterns reinforce the size and market-depth interpretation, but they are weaker than the characteristic-level regressions in Table 2.

6.6 Figure 5: Price informativeness over time

Read: Figure 5 tracks the cross-sectional mean, median, and standard deviation from 1985 to 2017.

- Median informativeness rises from about 1.66% to 2.95%.
- Mean informativeness rises from about 1.72% to 3.29%.
- Dispersion also rises.
- Dispersion spikes around the early 2000s and the 2008 financial crisis.

Interpretation. Prices become more informative on average, but informativeness also becomes more unequal across stocks.

6.7 Figure 6: Time trends by characteristics

Read: Figure 6 splits stocks into top and bottom halves by size, turnover, value, idiosyncratic volatility, institutional ownership, and analyst coverage.

- The gap between large and small stocks increases over time.
- A similar divergence appears for high versus low turnover and high versus low analyst coverage.
- Value and institutional ownership groups evolve more in parallel.
- Sorting by idiosyncratic volatility does not create a systematic difference over time.

Why it matters. The rise in average informativeness is not uniform. Large and actively traded stocks increasingly pull away from smaller and less liquid stocks.

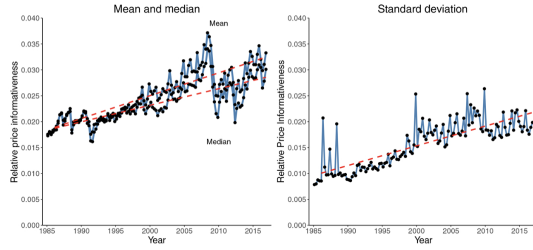


Figure 5
Price informativeness over time: Aggregate results
The left panel in Figure 5 shows the time-series evolution of the cross-sectional mean and median relative price informativeness. The right panel in Figure 5 shows the time-series evolution of the cross-sectional standard deviation of price informativeness. The red dashed lines represent linear trends. In both panels, the dots correspond to the average within a quarter of the price informativeness measures computed using year-over-year changes and overlapping quarterly data.

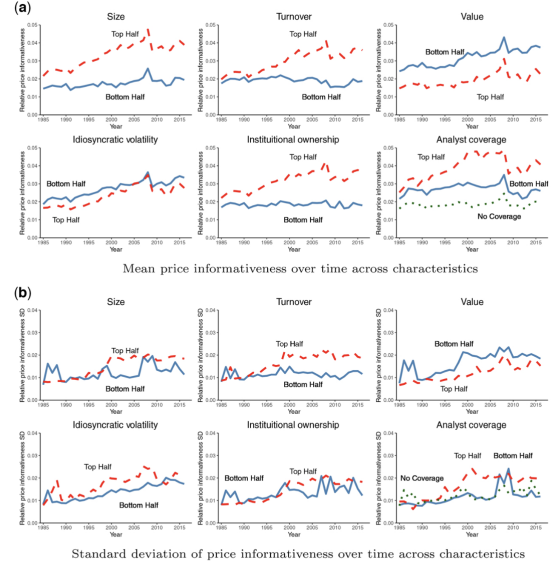


Figure 6
Price informativeness over time: Cross-sectional characteristics
Panels a and b show the time-series evolution of the mean relative price informativeness (panel a) and its standard deviation (panel b) for the top and bottom of the distribution of price informativeness for each characteristic (size, turnover, value, idiosyncratic volatility, institutional ownership, and analyst coverage). The red dashed lines correspond to the top half of the distribution, while the solid blue lines correspond to the bottom half of the distribution. The green dotted line in the last panel corresponds to stocks with no analyst coverage. In this figure, stocks are assigned to each half every quarter and observations are aggregated at the annual level.

6.8 Figures 7 and 8: Unlearnable component results

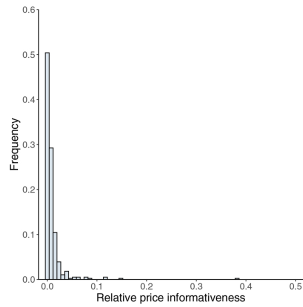


Figure 7
Price informativeness about learnable component: Relative-frequency histogram
This figure shows a relative-frequency histogram of price informativeness about the learnable component of future earnings for a representative time period, the last quarter of 2012. The histogram features 383 stocks.

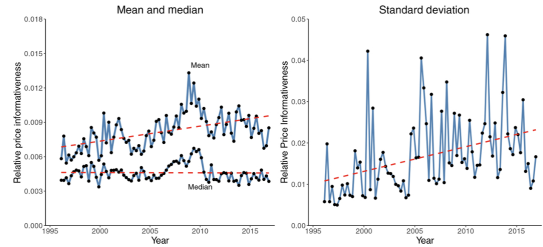


Figure 8
Price informativeness about learnable component over time: Aggregate results
The left panel shows the time-series evolution of the cross-sectional mean and median relative price informativeness about the learnable component of future earnings. The right panel shows the time-series evolution of the cross-sectional standard deviation of price informativeness. The red dashed lines represent linear trends starting in 1996. In both panels, the dots correspond to the average within a quarter of the price informativeness measures computed using year-over-year changes and overlapping quarterly data.

Read: Figures 7 and 8 implement the model in which only part of earnings is learnable, proxied by analyst forecasts.

- In the last quarter of 2012, the unlearnable-component model gives a median informativeness of 0.41%, compared with 2.31% in the baseline.
- From 1996 to 2017, average median informativeness is 0.47% and average mean informativeness is 0.85%.
- Mean and dispersion increase over time, while the median is nearly flat.
- The correlation between baseline informativeness and learnable-component informativeness is only 0.064.

Table 3
Distribution of average price informativeness at different horizons

Tenor	Mean	Std. Dev	P5	P95	N
1	0.02614	0.01745	0.005057	0.0556	201,511
2	0.00805	0.00957	0.000787	0.0218	153,287
3	0.00928	0.01761	0.000305	0.0326	9,231
4	0.01087	0.01988	0.000368	0.0340	9,800
5	0.00472	0.00730	0.000361	0.0129	100,259

This table shows results for price informativeness about growth rates 1, 2, 3, 4, and 5 years ahead. The table reports the mean, the standard deviation and the 5th and 95th percentiles of average price informativeness for the different horizons.

Interpretation. Once the paper asks only about the part of earnings analysts can plausibly forecast, estimated informativeness is much lower. This suggests that payoff measurement matters a lot.

6.9 Table 3: Longer horizons

Read: Table 3 estimates informativeness about payoff growth from one to five years ahead.

Horizon	Mean	SD	P5	P95	N
1 year	0.02614	0.01745	0.005057	0.0556	201,511
2 years	0.00805	0.00957	0.000787	0.0218	153,287
3 years	0.00928	0.01761	0.000305	0.0326	9,231
4 years	0.01087	0.01988	0.000368	0.0340	9,800
5 years	0.00472	0.00730	0.000361	0.0129	100,259

Interpretation.

- Informativeness is highest at the one-year horizon.
- It generally declines as the horizon lengthens.
- The 3- and 4-year estimates have fewer observations, so the small increase from 2 to 4 years should not be overinterpreted.

7 Relation to existing informativeness measures

7.1 Posterior variance and forecasting price efficiency

The paper compares relative price informativeness with two related objects:

- **Posterior variance:** $VP = \text{Var}[u_t \mid \Delta p_t, \Delta x_t, \Delta \chi_t]$.
- **Forecasting price efficiency:** $V_{\text{FPE}} = \text{Var}[\mathbb{E}[u_t \mid \Delta p_t, \Delta x_t, \Delta \chi_t] \mid \Delta x_t, \Delta \chi_t]$.

By the law of total variance,

$$\text{Var}[u_t \mid \Delta x_t, \Delta \chi_t] = \mathbb{E}[\text{Var}[u_t \mid \Delta p_t, \Delta x_t, \Delta \chi_t] \mid \Delta x_t, \Delta \chi_t] + \text{Var}[\mathbb{E}[u_t \mid \Delta p_t, \Delta x_t, \Delta \chi_t] \mid \Delta x_t, \Delta \chi_t].$$

When uncertainty is Gaussian,

$$VP = \frac{1 - \tau_\pi^R}{\tau_{u|\Delta\chi_t}}, \quad V_{\text{FPE}} = \frac{\tau_\pi^R}{\tau_{u|\Delta\chi_t}}.$$

Interpretation.

- VP and V_{FPE} depend on payoff volatility.
- τ_{π}^R is unit-free and therefore better suited for comparing informativeness across assets.
- Measures like forecasting price efficiency can change because payoff volatility changes, even if the information content of prices does not.

7.2 Informativeness versus predictability

- Price informativeness and return/payout predictability are related but not identical.
- The paper regresses prices on future payoffs to recover how much information prices contain.
- This is different from forecasting regressions that regress future payoffs or returns on current prices.
- The right empirical specification depends on the theoretical object being identified.

8 Limitations and open issues

- **Linearity.** The exact identification formula is derived under a linear or log-linear pricing equation. Nonlinear information aggregation may require different formulas.
- **Feedback effects.** The baseline framework abstracts from feedback from prices to fundamentals. If prices affect investment, financing, or real decisions, payoff innovations may not be exogenous to prices.
- **Discount rates.** The reduced-form pricing equation can include risk premia as noise, but a richer model is needed to separate information from discount-rate variation.
- **Payoff measurement.** Earnings are a practical payoff proxy, but they are not the only possible fundamentals. Different payoff definitions could produce different informativeness estimates.
- **Learnable component.** Analyst forecasts are a noisy proxy for the learnable component of earnings and may themselves respond to prices.
- **Frequency.** Higher-frequency data could give more precise estimates, but high-frequency payoff measures are difficult to observe.
- **Welfare.** More informative prices may improve allocation, but welfare implications depend on the model and on whether information acquisition is socially excessive or beneficial.

9 Short summary

- The paper defines relative price informativeness as the fraction of uncertainty about future payoff innovation resolved by observing the price.
- The key identification result is

$$\tau_{\pi}^R = \frac{R_{\Delta x, \Delta x'}^2 - R_{\Delta x}^2}{1 - R_{\Delta x}^2}.$$

- The formula uses regressions of price changes on current and future payoff changes.
- The source of price noise does not need to be observed.
- The method applies to models with sentiment, representative-agent risk premia, informed/uninformed investors, and noise traders.
- Empirically, stock-specific price informativeness is small, right-skewed, and rising over time.
- Larger, more liquid, and more heavily followed stocks have more informative prices.
- When the authors focus on the learnable component of earnings, estimated informativeness is much lower.
- Informativeness is generally lower at longer horizons.

10 Conclusion

Core conclusion

Dávila and Parlato show that price informativeness can be identified from a normalized difference in R^2 across two regressions. This transforms a central theoretical object in information aggregation models into something that can be measured in the data.

Economic intuition

Prices reveal information because informed or better-motivated traders push prices toward future fundamentals. Prices are noisy because they also reflect sentiment, liquidity needs, risk-bearing capacity, or noise-trader demand. The paper’s insight is that one can recover the relative precision of the price signal without observing the noise directly.

Takeaway

The paper’s empirical estimates suggest that U.S. stock prices have become more informative over time, especially for large and actively traded stocks. But the estimates remain modest and depend importantly on the definition of payoffs and the learnable component of fundamentals.